

SMAPhysics.com

JEE Main Physics Practice Sheet — DPP-038

Topic: Potential Energy, Gravitational PE & Relation between Force and PE ($\vec{F} = -\vec{\nabla}U$)

Time Allowed: 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

Q1. The potential energy of a particle in a one-dimensional conservative force field is given by $U(x) = (2x^3 - 9x^2 + 12x)$ J, where x is in meters. The positions of stable and unstable equilibrium are respectively:

- (A) $x = 1$ m and $x = 2$ m
- (B) $x = 2$ m and $x = 1$ m
- (C) $x = 3$ m and $x = 0$ m
- (D) $x = 0$ m and $x = 3$ m

Q2. The potential energy function for a two-dimensional force field is $U(x, y) = (3x^2y - 4y^3)$ J. The force vector $\vec{F}$ acting on a particle at the point (1 m, 1 m) is:

- (A) $(-6\hat{i} + 9\hat{j})$ N
- (B) $(6\hat{i} - 9\hat{j})$ N
- (C) $(-6\hat{i} - 9\hat{j})$ N
- (D) $(9\hat{i} - 6\hat{j})$ N

Q3. A uniform chain of mass M and length L is held on a smooth horizontal table such that a length y hangs vertically over the edge. Taking the tabletop as the reference level ($U = 0$), the gravitational potential energy of the chain is:

- (A) $-\frac{Mgy^2}{2L}$
- (B) $-\frac{Mgy}{L}$
- (C) $+\frac{Mgy^2}{2L}$
- (D) $-\frac{Mg(L-y)^2}{2L}$

Q4. A conservative force $\vec{F} = -\frac{k}{r^2}\hat{r}$ acts on a particle. If the potential energy is chosen to be zero at infinity ($U(\infty) = 0$), the potential energy at distance r from the origin is:

- (A) $+\frac{k}{r}$
- (B) $-\frac{k}{2r^2}$
- (C) $-\frac{k}{r}$
- (D) $+\frac{k}{2r}$

Q5. A spherical solid planet of mass M and radius R creates a gravitational potential energy field for an external particle of mass m . The work done by gravity in moving the particle from $r = R$ to $r = 3R$ is:

- (A) $-\frac{2GMm}{3R}$
- (B) $+\frac{2GMm}{3R}$

- (C) $+\frac{GMm}{3R}$
- (D) $-\frac{GMm}{3R}$

Q6. The potential energy of a diatomic molecule as a function of the interatomic distance r is given by $U(r) = \frac{A}{r^{12}} - \frac{B}{r^6}$ ($A, B > 0$). The equilibrium bond length r_0 is:

- (A) $\left(\frac{A}{B}\right)^{1/6}$
- (B) $\left(\frac{2A}{B}\right)^{1/6}$
- (C) $\left(\frac{B}{2A}\right)^{1/6}$
- (D) $\left(\frac{A}{2B}\right)^{1/6}$

Q7. In Question 6, the dissociation energy (binding energy) of the diatomic molecule is:

- (A) $\frac{B^2}{4A}$
- (B) $\frac{B^2}{2A}$
- (C) $\frac{A^2}{4B}$
- (D) $\frac{B^2}{8A}$

Q8. The relation between a conservative force $\vec{F}$ and its potential energy U in vector form is:

- (A) $\vec{F} = \vec{\nabla}U$
- (B) $\vec{F} = -\vec{\nabla}U$
- (C) $\vec{F} = \vec{\nabla} \times U$
- (D) $U = -\vec{\nabla} \cdot \vec{F}$

Q9. A particle of mass $m = 1$ kg is constrained to move along the x -axis in a potential field $U(x) = (x^2 - 4x + 7)$ J. If the total mechanical energy of the particle is $E = 12$ J, the maximum speed of the particle is:

- (A) 3.0 m/s
- (B) 4.24 m/s
- (C) 5.0 m/s
- (D) 2.0 m/s

Q10. An elastic sphere of mass m is compressed between two fingers so that its potential energy increases by ΔU . If this system is placed in an elevator accelerating upward at a , the change in its internal elastic potential energy:

- (A) Increases by $m(g+a)h$

- (B) Remains completely unchanged (ΔU)
 (C) Decreases by $\frac{1}{2}ma^2$
 (D) Becomes zero
- Q11.** If the potential energy function of a particle is $U(x) = \alpha x^4$ ($\alpha > 0$), the nature of equilibrium at $x = 0$ is:
 (A) Unstable equilibrium
 (B) Neutral equilibrium
 (C) Stable equilibrium
 (D) Not an equilibrium point
- Q12.** A particle moves in a force field where $U(x, y, z) = (2x + 3y - z)$ J. The magnitude of the force acting on the particle is:
 (A) $\sqrt{14}$ N
 (B) 6 N
 (C) 14 N
 (D) 4 N
- Q13.** A uniform rod of mass M and length L initially lying flat on the ground is tilted upright until it stands vertically. The change in gravitational potential energy of the rod is:
 (A) MgL
 (B) $\frac{1}{2}MgL$
 (C) $\frac{1}{3}MgL$
 (D) $\frac{1}{4}MgL$
- Q14.** A body moves in the xy -plane subject to a force $\vec{F} = (2xy\hat{i} + x^2\hat{j})$ N. The corresponding potential energy function $U(x, y)$ (taking $U(0, 0) = 0$) is:
 (A) $+x^2y$
 (B) $-x^2y$
 (C) $-2x^2y$
 (D) $+2xy^2$
- Q15.** A particle of mass m is displaced along the x -axis in a potential $U(x) = U_0(1 - \cos kx)$. For very small oscillations near $x = 0$, the time period of oscillation is:
 (A) $2\pi\sqrt{\frac{m}{U_0k^2}}$
 (B) $2\pi\sqrt{\frac{m}{U_0k}}$
 (C) $2\pi\sqrt{\frac{U_0}{mk^2}}$
 (D) $\frac{2\pi}{k}\sqrt{\frac{m}{U_0}}$
- Q16.** A force field is given by $\vec{F} = (y\hat{i} - x\hat{j})$ N. Is this force conservative, and does a potential energy function exist?
 (A) Yes, conservative; $U = -xy$
 (B) No, non-conservative; no scalar potential energy function exists
 (C) Yes, conservative; $U = \frac{1}{2}(x^2 + y^2)$
 (D) Yes, conservative; $U = \frac{1}{2}(y^2 - x^2)$
- Q17.** A body of mass m is raised from the surface of Earth (mass M_E , radius R_E) to an altitude $h = R_E$. The gain in gravitational potential energy is:
 (A) mgh
 (B) $\frac{1}{2}mgR_E$
 (C) $2mgR_E$
 (D) $\frac{1}{4}mgR_E$
- Q18.** The potential energy of a particle is given by $U(r) = \alpha r^3$ ($\alpha > 0$). The force $\vec{F}$ acting on the particle is:
 (A) $-3\alpha r^2\hat{r}$
 (B) $+3\alpha r^2\hat{r}$
 (C) $-\alpha r^2\hat{r}$
 (D) $-3\alpha r\hat{r}$
- Q19.** The potential energy curve of a particle shows a local maximum at position $x = x_0$. If the particle is placed at $x = x_0$ with zero velocity, then:
 (A) $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} > 0$ (Stable equilibrium)
 (B) $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} < 0$ (Unstable equilibrium)
 (C) $\frac{dU}{dx} \neq 0$ (Non-equilibrium)
 (D) $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} = 0$ (Neutral equilibrium)
- Q20.** A heavy rope of mass M and length L hangs vertically from a rigid ceiling. The gravitational potential energy of the rope relative to the ceiling ($U = 0$) is:
 (A) $-\frac{1}{2}MgL$
 (B) $-MgL$
 (C) $+\frac{1}{2}MgL$
 (D) $-\frac{1}{3}MgL$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-038)

Potential Energy, Gravitational PE & Relation between Force and PE

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	C	16	B
2	A	7	A	12	A	17	B
3	A	8	B	13	B	18	A
4	C	9	B	14	B	19	B
5	A	10	B	15	A	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)

Equilibrium condition: $F = -\frac{dU}{dx} = 0$.

$$\frac{dU}{dx} = 6x^2 - 18x + 12 = 0 \implies x = 1 \text{ m}, x = 2 \text{ m}$$

Second derivative: $\frac{d^2U}{dx^2} = 12x - 18$.

• At $x = 1$: $\frac{d^2U}{dx^2} = -6 < 0$ (Unstable).

• At $x = 2$: $\frac{d^2U}{dx^2} = +6 > 0$ (Stable).

Sol 2. Option (A)

Components of conservative force:

$$F_x = -\frac{\partial U}{\partial x} = -6xy$$

$$F_y = -\frac{\partial U}{\partial y} = -(3x^2 - 12y^2)$$

At (1, 1): $F_x = -6 \text{ N}, F_y = +9 \text{ N} \implies \vec{F} = (-6\hat{i} + 9\hat{j}) \text{ N}$.

Sol 3. Option (A)

Mass of hanging segment: $m_h = \frac{M}{L}y$.

Its center of mass lies at $y_{\text{cm}} = \frac{y}{2}$.

$$U = m_h g y_{\text{cm}} = \left(\frac{M}{L}y\right)g\left(\frac{y}{2}\right) = -\frac{Mgy^2}{2L}$$

Sol 4. Option (C)

Potential energy:

$$U(r) = -\int_{\infty}^r \vec{F} \cdot d\vec{r} = -\int_{\infty}^r \left(-\frac{k}{r^2}\right) dr = -\frac{k}{r}$$

Sol 5. Option (A)

Work done by gravity:

$$W_g = -\Delta U = -\left(-\frac{GMm}{3R} - \left(-\frac{GMm}{R}\right)\right) = -\frac{2GMm}{3R}$$

Sol 6. Option (B)

At equilibrium, $\frac{dU}{dr} = 0$:

$$-\frac{12A}{r^{13}} + \frac{6B}{r^7} = 0 \implies r^6 = \frac{2A}{B} \implies r_0 = \left(\frac{2A}{B}\right)^{1/6}$$

Sol 7. Option (A)

Dissociation energy $D = U(\infty) - U(r_0)$:

$$U(r_0) = \frac{A}{(2A/B)^2} - \frac{B}{(2A/B)} = -\frac{B^2}{4A}$$

$$D = 0 - \left(-\frac{B^2}{4A}\right) = \frac{B^2}{4A}$$

Sol 8. Option (B)

A conservative force is the negative gradient of potential energy: $\vec{F} = -\vec{\nabla}U$.

Sol 9. Option (B)

Minimum potential energy occurs at $x = 2 \text{ m}$:

$$U_{\text{min}} = U(2) = 2^2 - 4(2) + 7 = 3 \text{ J}$$

Maximum kinetic energy:

$$K_{\text{max}} = E - U_{\text{min}} = 12 - 3 = 9 \text{ J}$$

$$\frac{1}{2}(1)v_{\text{max}}^2 = 9 \implies v_{\text{max}} = \sqrt{18} \approx 4.24 \text{ m/s}$$

Sol 10. Option (B)

Internal elastic potential energy depends strictly on relative molecular compression, remaining invariant under overall acceleration.

Sol 11. Option (C)

$$F = -\frac{dU}{dx} = -4\alpha x^3.$$

For any displacement from $x = 0$, the force opposes the displacement (restoring force). Thus $x = 0$ is a **stable equilibrium**.

Sol 12. Option (A)

$$\vec{F} = -\vec{\nabla}U = -(2\hat{i} + 3\hat{j} - \hat{k}) \text{ N}$$

$$|\vec{F}| = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{14} \text{ N}$$

Sol 13. Option (B)

Initial center of mass is at $y_i = 0$; final is at $y_f = L/2$.

$$\Delta U = Mg\left(\frac{L}{2} - 0\right) = \frac{1}{2}MgL$$

Sol 14. Option (B)

$$-\vec{\nabla}U = \vec{F} \implies \frac{\partial U}{\partial x} = -2xy \text{ and } \frac{\partial U}{\partial y} = -x^2.$$

Integrating gives $U(x, y) = -x^2y$.

Sol 15. Option (A)

For small x , $\cos kx \approx 1 - \frac{k^2x^2}{2}$:

$$U(x) \approx \frac{1}{2}(U_0k^2)x^2 \implies k_{\text{eff}} = U_0k^2$$

$$T = 2\pi\sqrt{\frac{m}{U_0 k^2}}$$

Sol 16. Option (B)

Evaluating curl:

$$\vec{\nabla} \times \vec{F} = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \hat{k} = (-1 - 1)\hat{k} = -2\hat{k} \neq \vec{0}$$

Since the curl is non-zero, the force is non-conservative and no potential energy function exists.

Sol 17. Option (B)

$$\Delta U = U(2R_E) - U(R_E) = \frac{GM_E m}{2R_E} = \frac{1}{2}mgR_E$$

Sol 18. Option (A)

$$\vec{F} = -\frac{dU}{dr}\hat{r} = -\frac{d}{dr}(\alpha r^3)\hat{r} = -3\alpha r^2\hat{r}$$

Sol 19. Option (B)

At a local maximum of $U(x)$:

$$\frac{dU}{dx} = 0 \quad (\text{Equilibrium}), \quad \frac{d^2U}{dx^2} < 0 \quad (\text{Unstable})$$

Sol 20. Option (A)

Center of mass of the hanging rope is at $y_{\text{cm}} = -L/2$.

$$U = Mg \left(-\frac{L}{2} \right) = -\frac{1}{2}MgL$$